

Supplementary Information for:

Surface hydroxyl-induced Pt⁰ clusters on TiO₂ for synergistic water gas shift catalysis

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Supplementary Methods:

Transmission electron microscopy (TEM): The low-magnification TEM images were taken from a JEOL JEM-1011 microscope operated at 100 kV.

The X-ray diffraction (XRD): XRD pattern was conducted using a PANalytical X'Pert3 powder diffractometer operating at 40 kV and 40 mA, with Cu K α radiation ($\lambda = 0.15418$ nm).

Attenuated total reflectance-infrared spectroscopy (ATR-IR): ATR-FTIR analysis was performed using a Thermo-Nicolet iS50 FT-IR spectrometer. The spectral data were acquired over the range of 400–4000 cm^{-1} , utilizing a KBr beam splitter.

Thermogravimetric analysis (TGA): TGA experiments were performed in a N_2 atmosphere from room temperature to 800 $^{\circ}\text{C}$ using a DSC3+ instrument.

The H–D exchange experiment: The H–D exchange experiment was carried out in a custom-built microreactor coupled with a mass spectrometer (Builder PCSA-1000). The sample (0.10 g) was placed in a quartz tube and pretreated under an argon atmosphere at 120 $^{\circ}\text{C}$ for 1 hour, after which it was cooled to 30 $^{\circ}\text{C}$. Following pretreatment, the sample was purged with a 10 vol.% D_2/Ar mixture at a flow rate of 30 mL/min and heated to 800 $^{\circ}\text{C}$ at a rate of 10 $^{\circ}\text{C}/\text{min}$, at which point the H–D exchange experiment commenced to monitor the changes in the HD signal ($m/z = 3$).

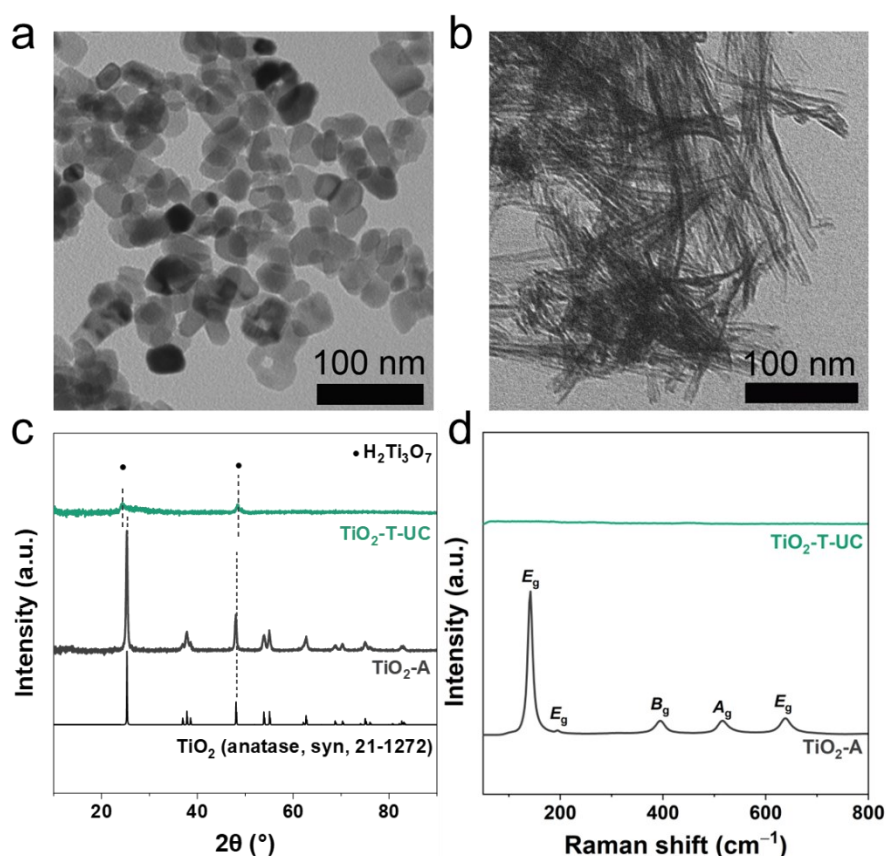
The surface reaction experiment monitored by mass spectrometry: All these experiments were performed on a lab-made reactor. The outlet gases were recorded by mass spectrum (LC-D200M, TILON). 50 mg catalysts were pretreated in 5% H_2/Ar at 300 $^{\circ}\text{C}$ for 0.5 h and then purging with Ar at room temperature. Subsequently, the catalyst was subjected to reaction under water-gas shift (WGS) reaction conditions. Then, H_2O dissociation experiment was conducted at 300 $^{\circ}\text{C}$ under $\text{H}_2\text{O}/\text{Ar}$ atmosphere. The catalysts were then heated under Ar atmosphere for 2h. Then, 2% CO/Ar atmosphere was injected, and signals were collected simultaneously.

The CO temperature programmed desorption (CO-TPD): All these experiments were performed on a lab-made reactor. The catalyst was first pretreated in a 5% H_2/Ar atmosphere for 30 minutes, then cooled to room temperature, followed by purging with the Ar atmosphere. Subsequently, the atmosphere was switched to 2% CO/Ar for 30 minutes. Then, Ar atmosphere

was switched again, followed by the purging process for 30 minutes. Subsequently, the temperature was ramped up to 800 °C at a rate of 2 °C/min while simultaneously collecting data.

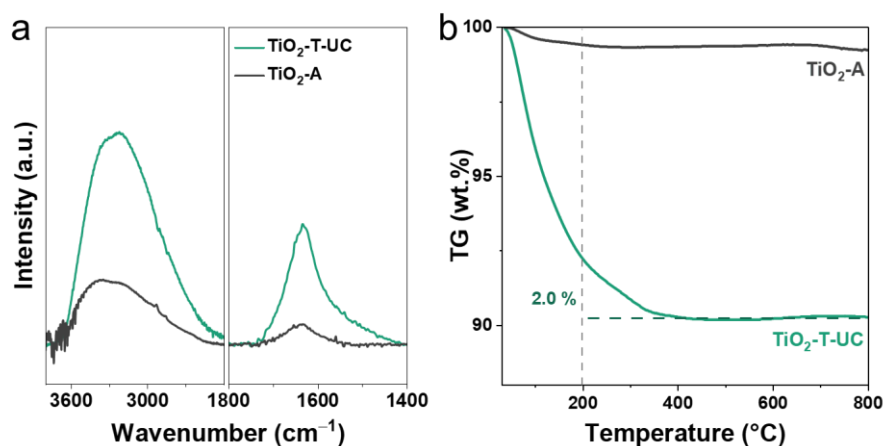
The Ar temperature programmed desorption (Ar-TPD): All these experiments were performed on a lab-made reactor. The catalyst was first pretreated in a 5% H₂/Ar atmosphere for 30 minutes, then cooled to room temperature, followed by purging with an Ar atmosphere for 30 minutes. Subsequently, the temperature was ramped up to 800 °C at a rate of 2 °C/min while simultaneously collecting data.

Supplementary Figures:



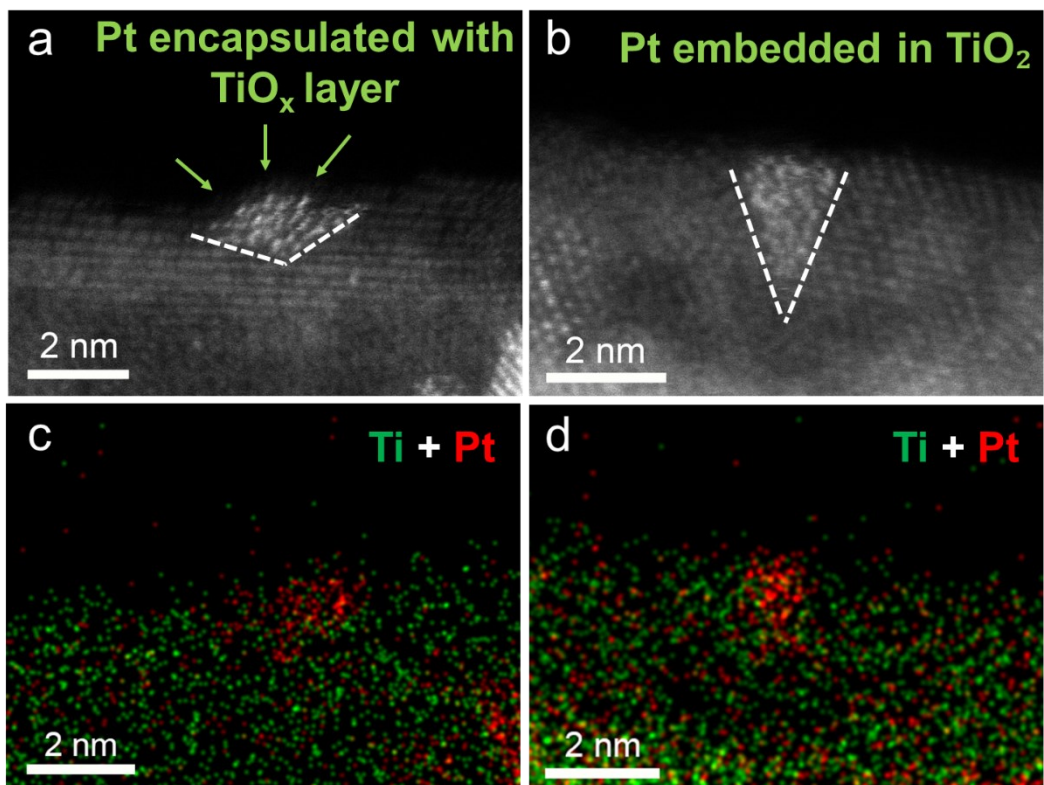
Supplementary Fig. S1 | The phase transformation from $\text{TiO}_2\text{-A}$ to $\text{TiO}_2\text{-T-UC}$. The transmission electron microscope (TEM) images of (a) $\text{TiO}_2\text{-A}$ and (b) $\text{TiO}_2\text{-T-UC}$. (c) XRD patterns of $\text{TiO}_2\text{-A}$ and $\text{TiO}_2\text{-T-UC}$. (d) Raman spectra of $\text{TiO}_2\text{-A}$ and $\text{TiO}_2\text{-T-UC}$.

By the hydrothermal method under highly concentrated NaOH (10 mol/L) and the following washing by HCl solution, the commercial anatase had entirely reconstructed into a nanotubular structure (Supplementary Fig. S1), which was characterized by amorphous $\text{H}_2\text{Ti}_3\text{O}_7$ phase (denoted as $\text{TiO}_2\text{-T-UC}$).



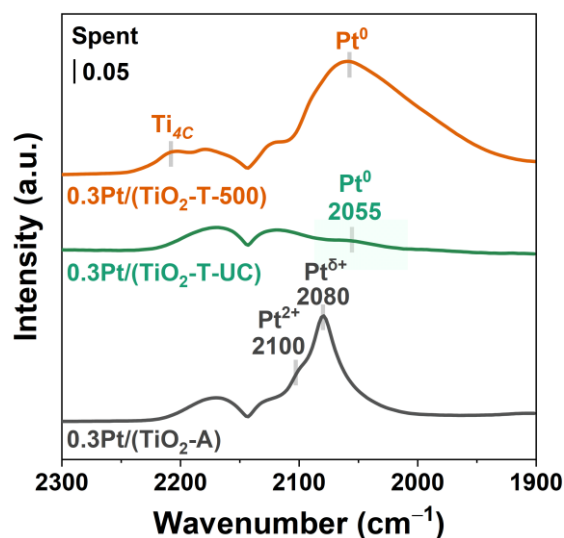
Supplementary Fig. S2 | The construction of hydroxyls on TiO₂-T-UC support. (a) The attenuated total reflectance-infrared spectroscopy (ATR-IR) and (b) TGA curves of TiO₂-T-UC and TiO₂-A.

Due to the phase transformation, a significant number of hydroxyls have been constructed on the TiO₂-T-UC surface, which was corroborated by the substantial increase in signals observed in the ATR-IR measurements of TiO₂-T-UC near 4000–2200 cm⁻¹ and 1800–1400 cm⁻¹ (Supplementary Fig. S2a). To further confirm the hydroxyl concentration on the TiO₂-T-UC supports, thermogravimetric analysis (TGA) was performed for quantitative assessment based on mass loss. The mass loss observed upon heating from room temperature to 200 °C often originated from the desorption of physically adsorbed H₂O, whereas that occurring above 200 °C was attributed to the desorption of hydroxyl groups (Supplementary Fig. S2b).



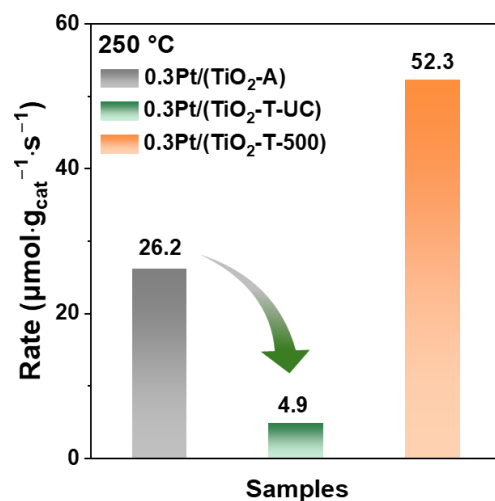
Supplementary Fig. S3 | The HAADF-STEM and EDS-mapping images of the spent 0.3Pt/(TiO₂-T-UC) catalyst. (a, b) Aberration-corrected HAADF-STEM images and (c, d) elemental mapping images of the spent 0.3Pt/(TiO₂-T-UC) catalyst.

For the spent 0.3Pt/(TiO₂-T-UC) catalyst, a portion of Pt species were semi-embedded in the surface of the support but were covered by a layer of TiO_x (Supplementary Fig. S3a), which was further evidenced by the overlapping positions of Pt and Ti elements in the EDS mapping images (Supplementary Fig. S3c). Another portion of the Pt species were completely embedded into the bulk of support oxide (Supplementary Fig. S3b and d).

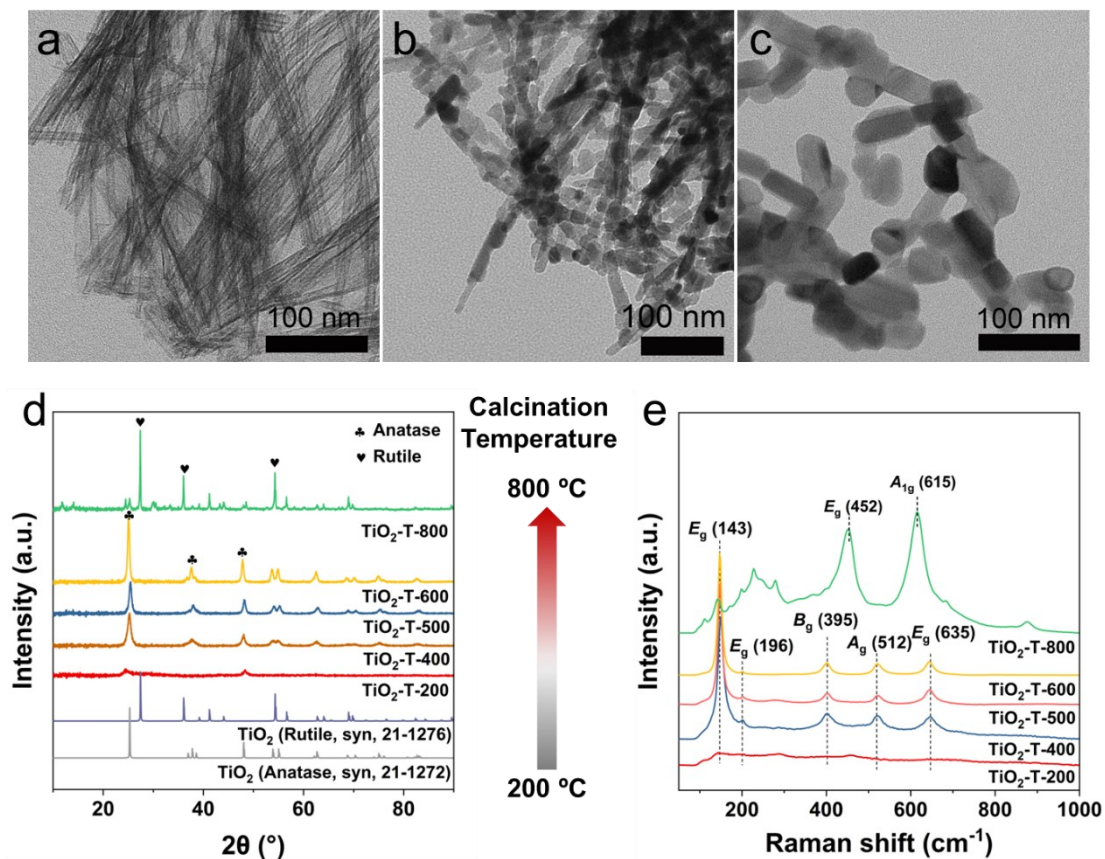


Supplementary Fig. S4 | The CO adsorption of Pt species in the spent catalyst. *In situ* DRIFTS spectra of the CO adsorption for the spent 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) samples.

To further investigate the exposure state of Pt species, *in situ* DRIFTS experiments using CO as the probe molecule were carried out (Supplementary Fig. S4). The 0.3Pt/(TiO₂-T-UC) catalyst showed almost no adsorption signal of CO on the Pt species, which further suggested that TiO₂-T-UC support was likely to have encapsulated the Pt species.

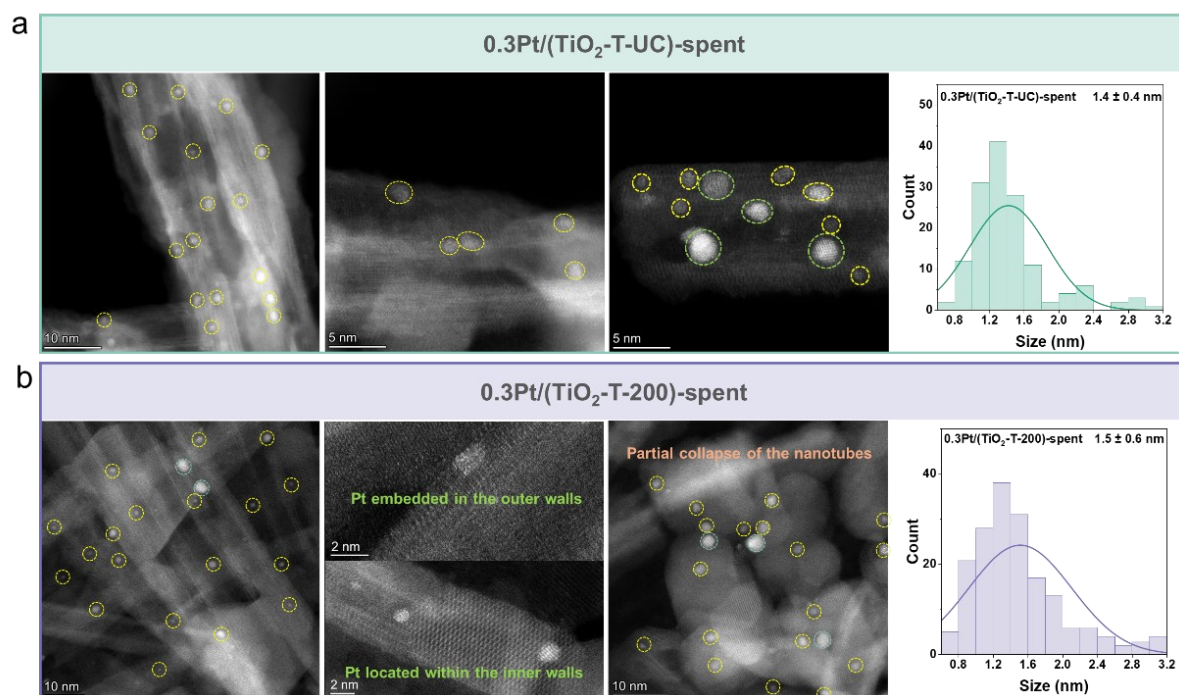


Supplementary Fig. S5 | The inhibited activity of the 0.3Pt/(TiO₂-T-UC) catalyst relative to the 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalysts. The reaction rate at 250 °C of 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) catalysts.



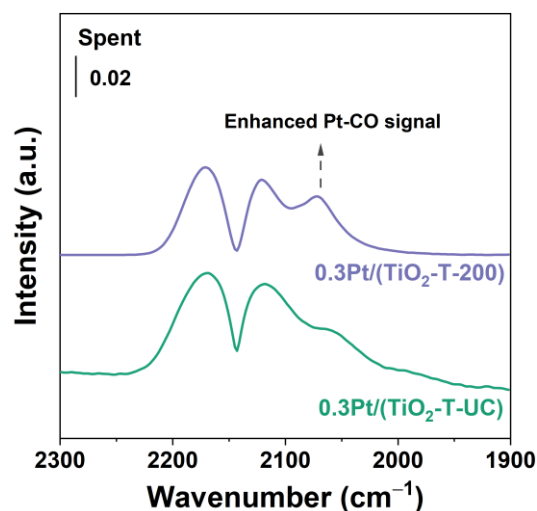
Supplementary Fig. S6 | Morphology and phase characterization of TiO_2 supports calcinated at various temperatures. The transmission electron microscope (TEM) image of (a) TiO_2 -T-200, (b) TiO_2 -T-500 and (c) TiO_2 -T-800. (d) XRD patterns and (e) Raman spectra of TiO_2 -T-200, TiO_2 -T-400, TiO_2 -T-500, TiO_2 -T-600 and TiO_2 -T-800.

Upon the further calcination of TiO_2 -T-UC, the nanotubes gradually collapsed to a short rod-like structure accompanied by recrystallization into anatase phase (TiO_2 -T- t_{cal}). At a higher temperature of calcination, the rutile phase appeared along with sintering.



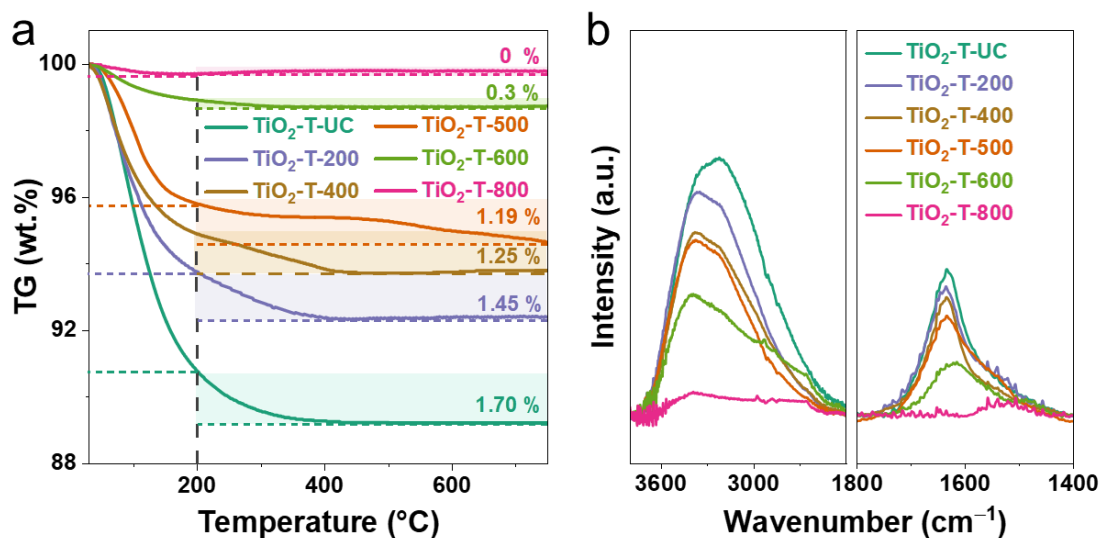
Supplementary Fig. S7 | The morphology and size distribution. The aberration-corrected HAADF-STEM images and the size statistics of the spent (a) 0.3Pt/(TiO₂-T-UC) and (b) 0.3Pt/(TiO₂-T-200) catalyst.

As shown in Supplementary Fig. S7a, the catalysts with supports that were untreated retained their nanotubular morphology after the WGS reaction. Interestingly, the Pt species in these samples were distributed both on the inner and outer walls of the nanotubes, resulting in the non-uniform size distribution. In addition to the inadequate exposure of active sites for Pt particles located within the nanotube interior, those on the external surfaces were also found to be embedded within the tube walls. These masked sites collectively contribute to the suppression of WGS activity. When the support was calcined at 200 °C, the majority of the nanotubes were retained (Supplementary Fig. S7b). Similarly to the 0.3Pt/(TiO₂-T-UC) catalyst, Pt particles were incorporated into both the interior and the outer walls of these nanotubes, which hindered the exposure of active sites. In contrast to 0.3Pt/(TiO₂-T-UC), a portion of the nanotubes in the 0.3Pt/(TiO₂-T-200) catalyst collapsed, forming layered sheet-like structures.



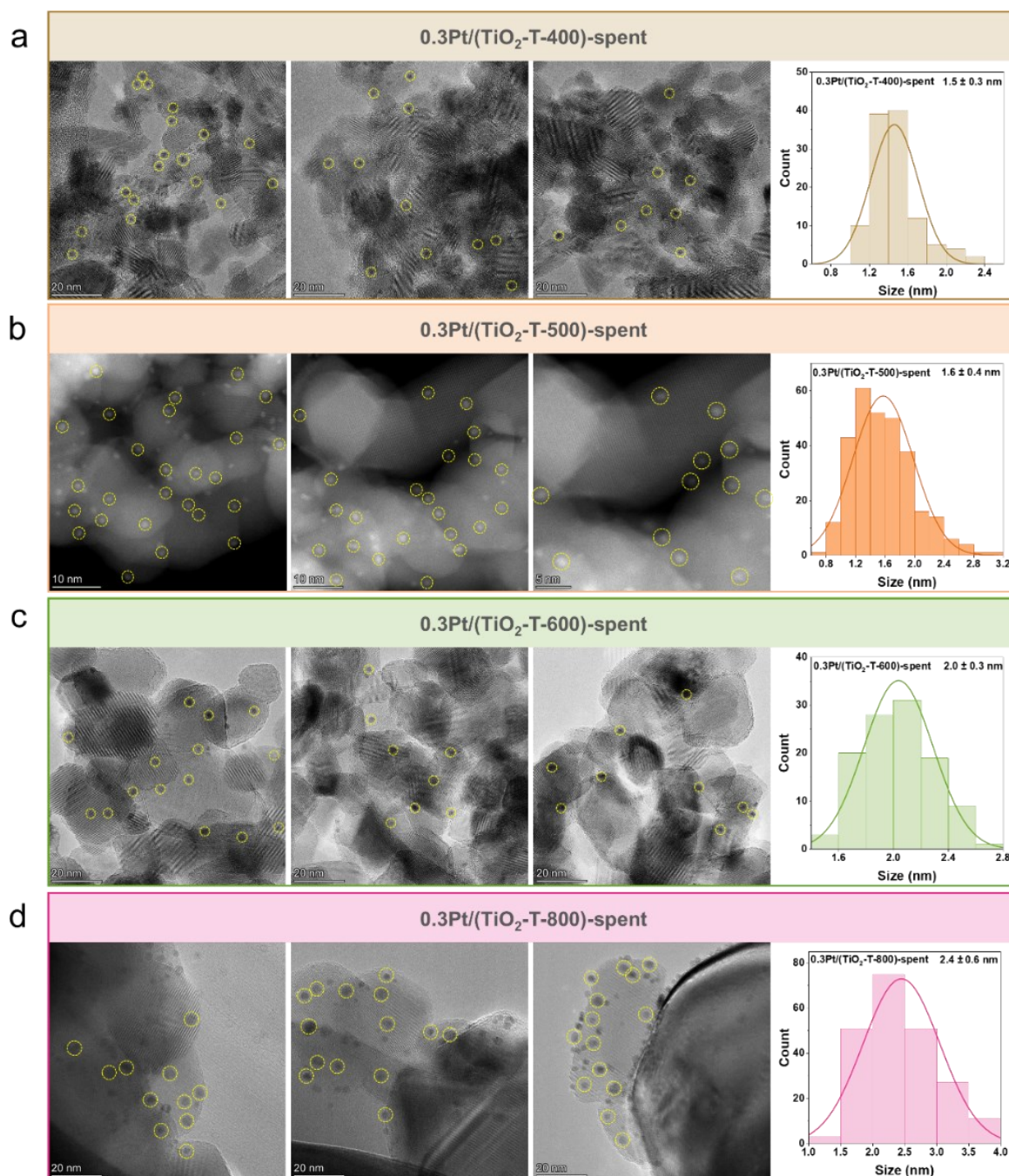
Supplementary Fig. S8 | The CO adsorption of Pt species in the spent 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-200) catalysts. *In situ* DRIFTS spectra of the CO adsorption for the spent 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-200) catalyst.

The partial collapse of the nanotubular structure in the 0.3Pt/(TiO₂-T-200) catalyst led to the exposure of a fraction of the Pt species, as evidenced by the enhanced CO adsorption intensity in the *in situ* DRIFTS spectra compared with that of 0.3Pt/(TiO₂-T-UC).



Supplementary Fig. S9 | Variation in surface hydroxyls content with increasing thermal treatment temperature. (a) TGA curves and (b) The ATR-IR spectra of various TiO₂ under different thermal treatment temperature.

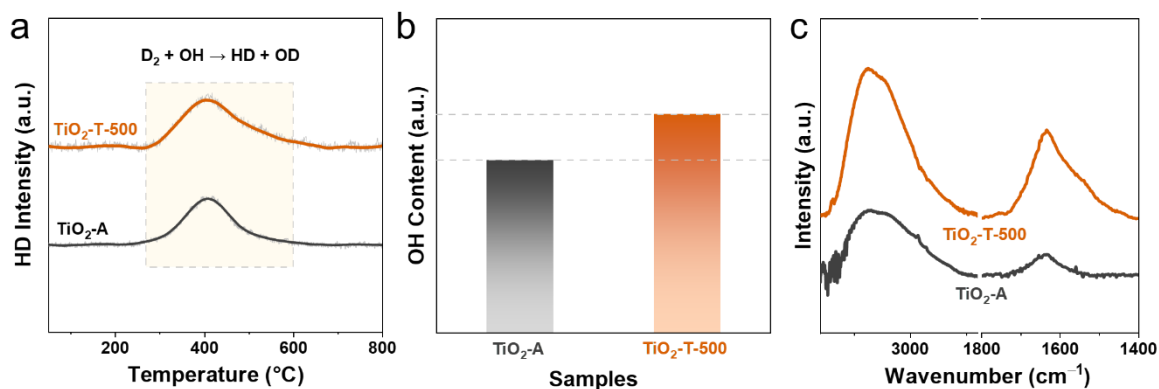
To further compare hydroxyl concentration on the three TiO₂ supports, thermogravimetric analysis (TGA) was performed for quantitative assessment based on mass loss, as the OH groups on the oxide surface were released as water molecules during the heating process. The mass loss observed upon heating from room temperature to 200 °C often originated from the desorption of physically adsorbed H₂O, whereas that occurring above 200 °C was attributed to the desorption of hydroxyl groups. As the heat treatment temperature increased, the mass loss decreased accordingly (Supplementary Fig. S9a). Subsequently, all supports were subjected to ATR-IR characterization (Supplementary Fig. S9b). It was observed that the intensity of the OH signal gradually decreased with increasing calcination temperature of the supports, further demonstrating that the calcination temperature plays a regulatory role in the surface hydroxyl content.



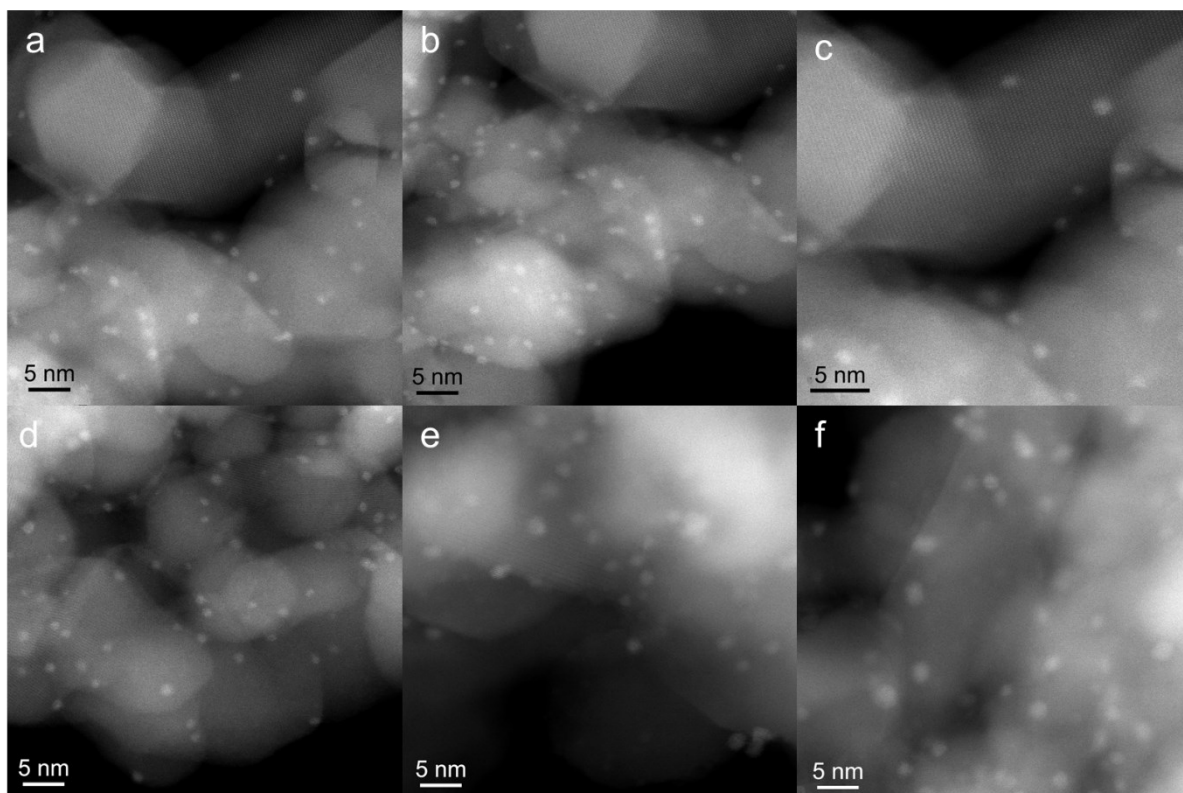
Supplementary Fig. S10 | The morphology and size distribution. The aberration-corrected HAADF-STEM images and the size statistics of the spent (a) 0.3Pt/(TiO₂-T-400), (b) 0.3Pt/(TiO₂-T-500), (c) 0.3Pt/(TiO₂-T-600) and (d) 0.3Pt/(TiO₂-T-800) catalyst.

As shown in Supplementary Fig. S10, when the calcination temperature of the support was increased to 400 °C, the TiO₂ nanotubes underwent complete collapse, transforming into the sheet-like morphology. This transformation was accompanied by a progressive increase in

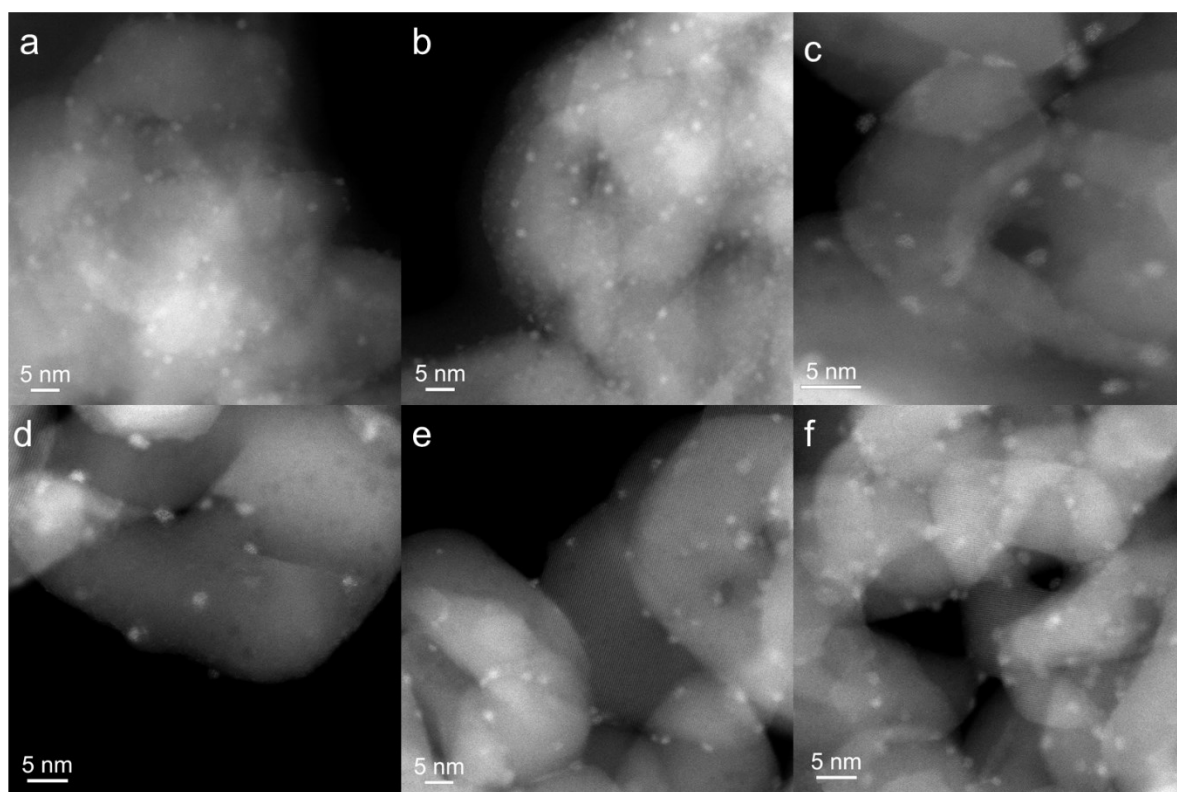
the size of the supported Pt particles with rising calcination temperature of TiO₂ support, which was attributed to the loss of surface hydroxyl groups and the consequent weakening of their anchoring effect on the Pt species.



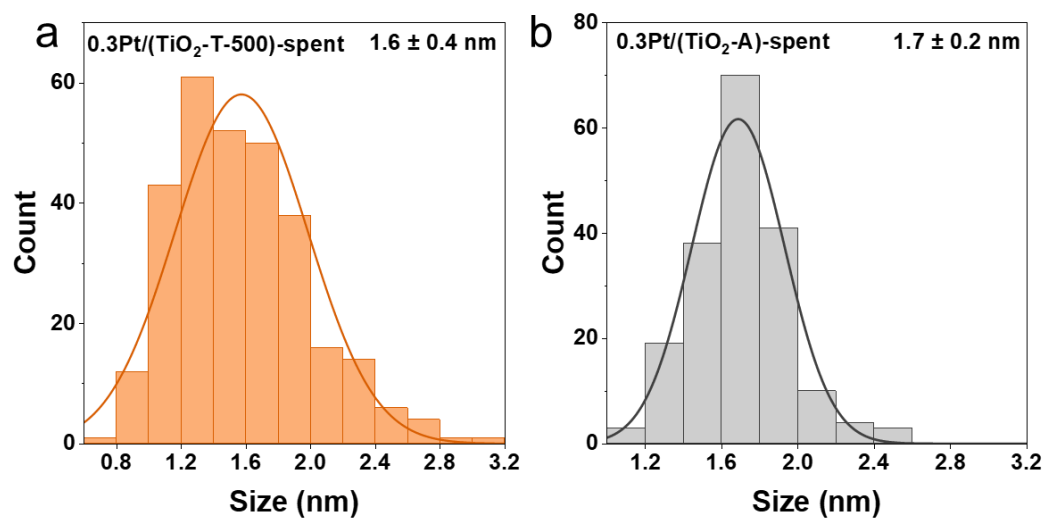
Supplementary Fig. S11 | Comparison of surface hydroxyls contents between $\text{TiO}_2\text{-T-500}$ and $\text{TiO}_2\text{-A}$. (a) HD evolution profiles during the H–D exchange of $\text{TiO}_2\text{-A}$ and $\text{TiO}_2\text{-T-500}$; (b) OH contents determined from the peak area of H–D curves normalized by support mass. (c) ATR-IR of $\text{TiO}_2\text{-A}$ and $\text{TiO}_2\text{-T-500}$ samples.



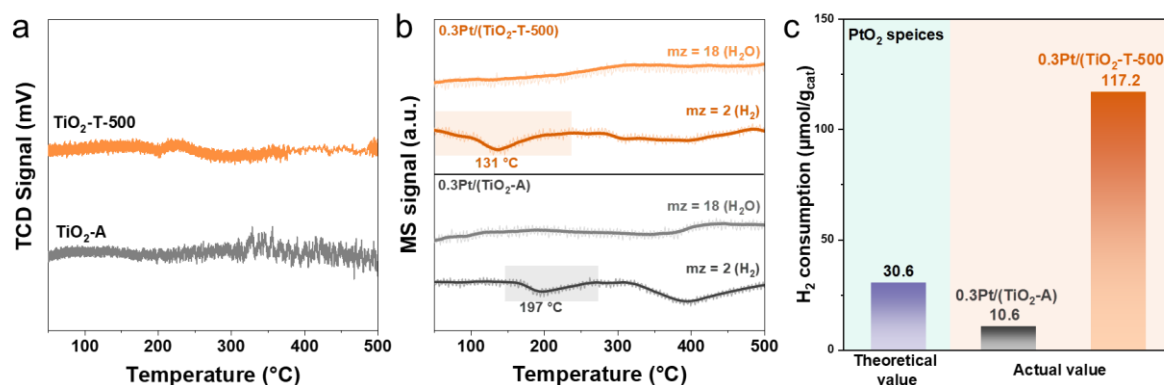
Supplementary Fig. S12 | The morphology of the spent 0.3Pt/(TiO₂-T-500) catalyst. (a–f) Aberration-corrected HAADF-STEM images of the spent 0.3Pt/(TiO₂-T-500).



Supplementary Fig. S13 | The morphology of the spent 0.3Pt/(TiO₂-A) catalyst. (a–f) Aberration-corrected HAADF-STEM images of the spent 0.3Pt/(TiO₂-A).

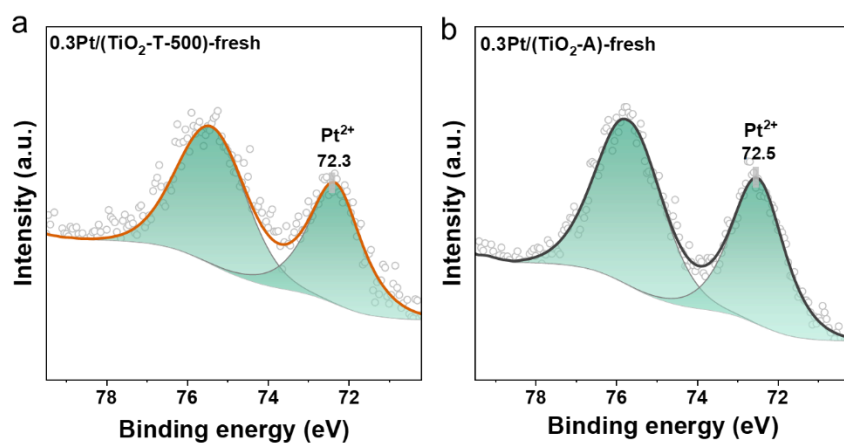


Supplementary Fig. S14 | The statistical analysis of Pt species. The size statistics of the spent (a) 0.3Pt/(TiO₂-T-500) and (b) 0.3Pt/(TiO₂-A) catalysts.

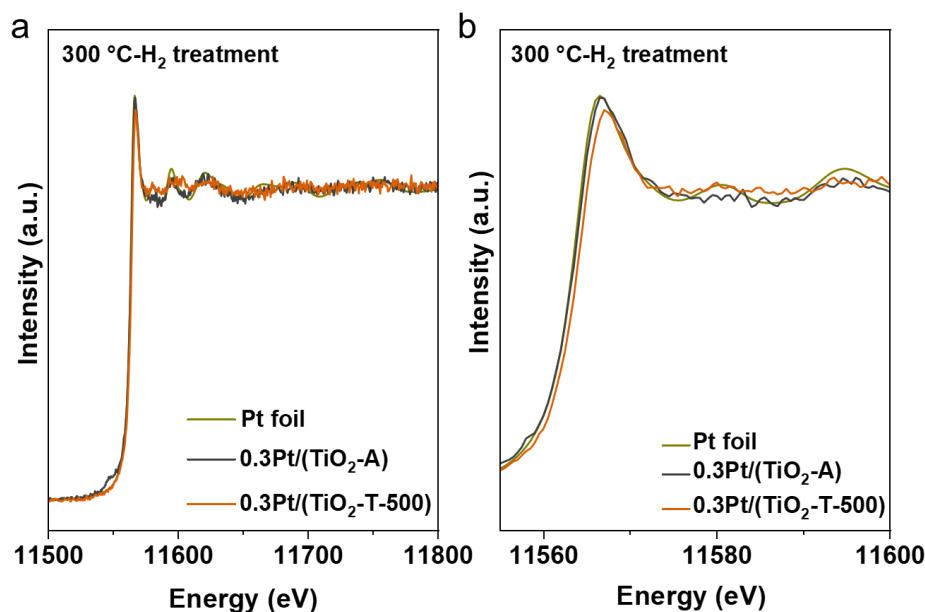


Supplementary Fig. S15 | The reduction of 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts. (a) H₂-TPR profiles of TiO₂-T-500 and TiO₂-A support. (b) H₂-TPR profiles of the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples characterized by mass spectrometry. (c) H₂ consumption calculated from the results in Figure 2a.

The TiO₂-T-500 and TiO₂-A supports exhibited negligible TCD signals during H₂-TPR analysis (Supplementary Fig. S15a), which indicated that the surface hydroxyl groups did not consume H₂, thereby ruling out the possibility that the lower reduction temperature was caused by dehydrogenation of hydroxyl groups. To exclude the potential influence of H₂O desorption during the heating process on the decrease in the reduction peak temperature, H₂-TPR coupled with mass spectrometry was performed on both catalysts (Supplementary Fig. S15b). The results indicated that the hydrogen consumption temperature of the 0.3Pt/(TiO₂-T-500) catalyst remained lower than that of 0.3Pt/(TiO₂-A). Moreover, no significant H₂O signal was detected simultaneously with the H₂ consumption peaks. This observation was likely attributable to the fact that both supports were calcined at 500 °C and subsequently the catalyst pretreated at 120 °C under the Ar atmosphere prior to the H₂-TPR measurements. The findings demonstrated that the lower reduction temperature did not arise from the water loss. Subsequently, we further quantified the H₂ consumption from the H₂-TPR results of the two catalysts (Supplementary Fig. S15c). It was found that the actual H₂ consumption of the 0.3Pt/(TiO₂-T-500) catalyst exceeded the theoretical H₂ consumption of the Pt species, whereas the actual H₂ consumption of 0.3Pt/(TiO₂-A) was less than its theoretical Pt consumption (Supplementary Fig. S15c). The result clearly demonstrated that the Pt species in the 0.3Pt/(TiO₂-T-500) catalyst activated the support, thereby leading to enhanced reducibility.

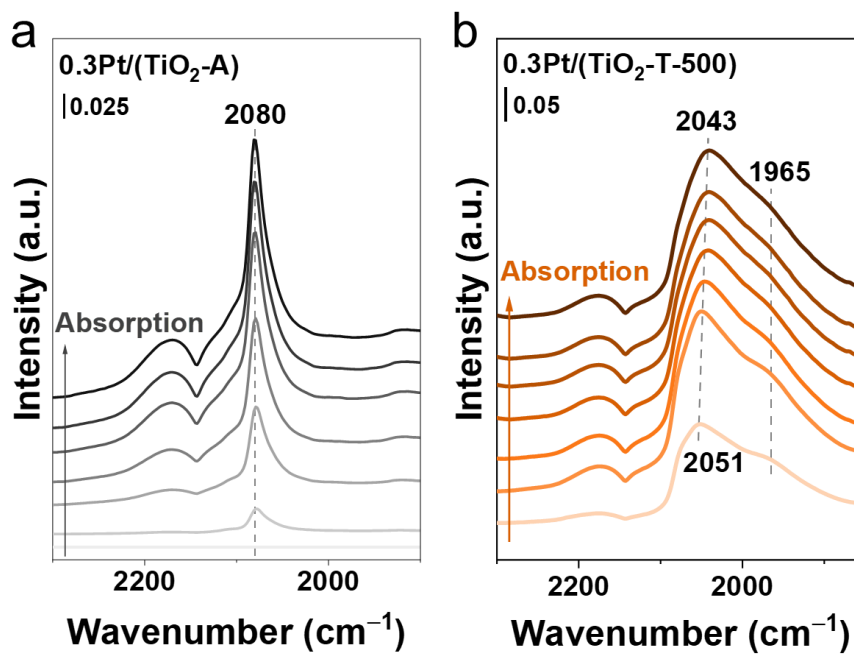


Supplementary Fig. S16 | The electronic structure of Pt species in the fresh 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts. The Pt 4f XPS for the fresh (a) 0.3Pt/(TiO₂-T-500) and (b) 0.3Pt/(TiO₂-A) samples.

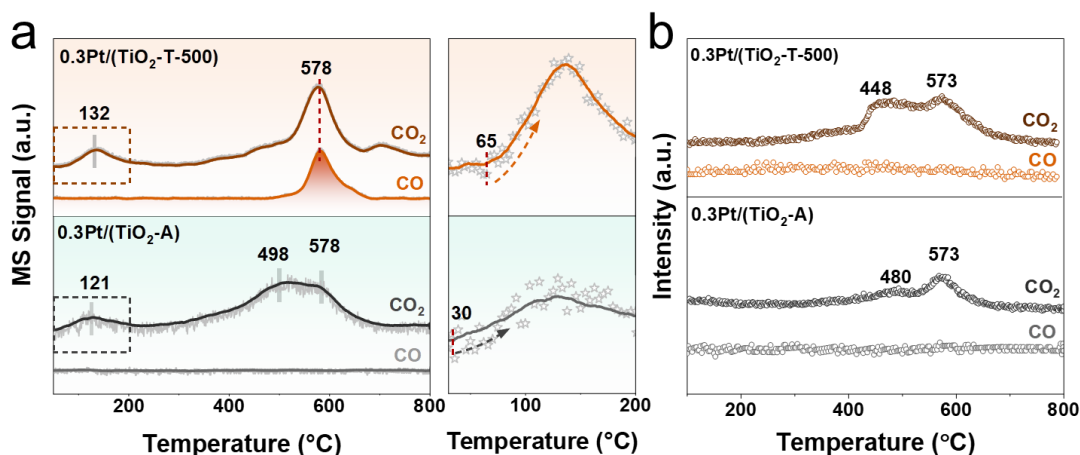


Supplementary Fig. S17 | The electronic state of Pt species in the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples under H₂ pretreatment. (a) XANES spectra of Pt L₃ edge for 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) samples catalysts under H₂ pretreatment at 300 °C. (b) Magnified view of panel (a).

Despite the relatively poor signal-to-noise ratio of the Pt–TiO₂ catalysts due to the low Pt loading (0.3 wt%), the difference in white-line intensity between the two catalysts under H₂ flow was still distinguishable. Compared with 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-500) exhibited a weaker white-line intensity, indicating that Pt species in 0.3Pt/(TiO₂-T-500) possessed higher 5d electron density. In addition, the Pt L₃-edge white-line intensity decreased and the edge shifted to higher energy relative to Pt foil, suggesting the H adsorption on Pt species modulated the Pt electronic state and decreased the unoccupied 5d states.⁵ Such hydrogen adsorption was also considered to be present in the 0.3Pt/(TiO₂-A) sample. This accounted for the consistently low white line intensity observed in both catalysts. Nevertheless, 0.3Pt/(TiO₂-T-500) exhibited an even lower white line intensity than 0.3Pt/(TiO₂-A), indicating the presence of a greater number of Pt⁰ sites.

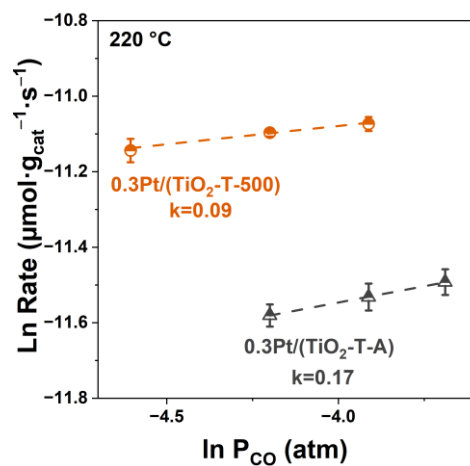


Supplementary Fig. S18 | Chemical state of Pt species in 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts during the WGS reaction. *In situ* DRIFTS experiments of (a) 0.3Pt/(TiO₂-A) and (b) 0.3Pt/(TiO₂-T-500) during the WGS reaction at 250 °C.

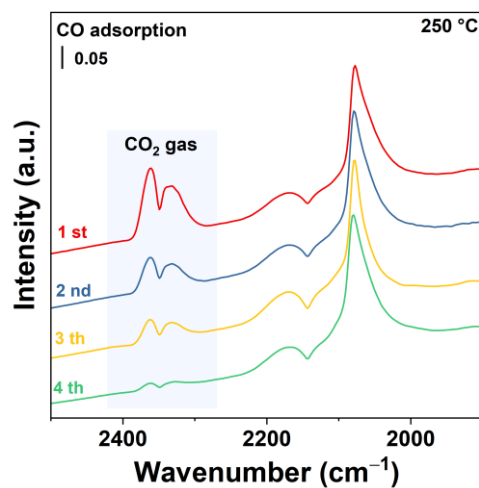


Supplementary Fig. S19 | CO adsorption strength in 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts. (a) CO-TPD of 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalyst. Right: the magnified CO₂ signal circled in the square within the range of 30 °C to 200 °C. (b) Ar-TPD of 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalyst.

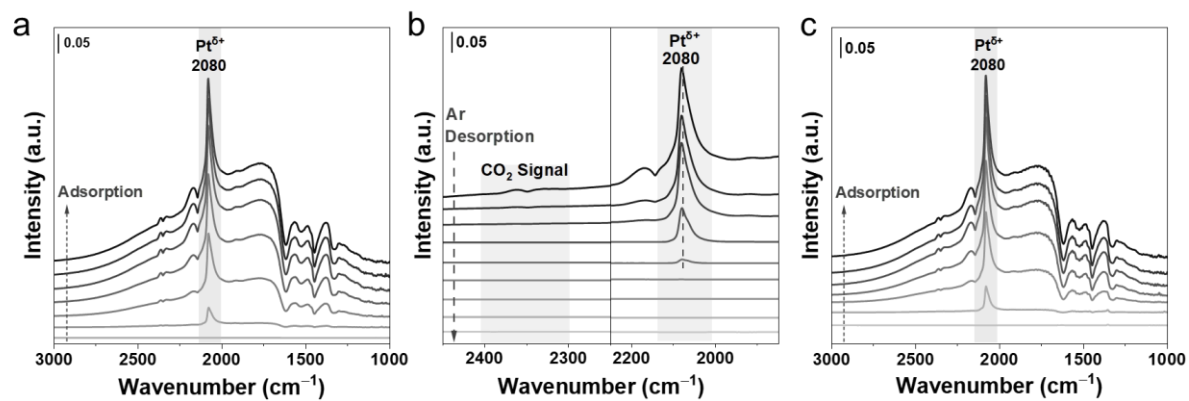
To evaluate the adsorption and activation capabilities of CO molecules, CO temperature-programmed desorption (CO-TPD) experiments were conducted, using Ar-TPD as references, to ensure the accuracy of the obtained results. For both samples, the broad desorption peaks of CO₂ between 400 °C and 700 °C observed in the Ar-TPD experiment was attributed to the decomposition of the surface carbonate species (Supplementary Fig. S19b). After H₂ reduction and pre-adsorption of CO molecules, the CO desorbed from the Pt species at low temperatures reacted with surface oxygen and was detected as the CO₂ signal (Supplementary Fig. S19a). For the 0.3Pt/(TiO₂-A) catalyst, CO₂ molecules occurred at 30 °C, which was significantly lower than the 65 °C observed for the 0.3Pt/(TiO₂-T-500) catalyst. In addition to the CO₂ signal, the 0.3Pt/(TiO₂-T-500) catalyst showed an additional CO desorption peak at 578 °C, indicating that the Pt species in the 0.3Pt/(TiO₂-T-500) sample exhibited significantly stronger CO binding compared to the 0.3Pt/(TiO₂-A) catalyst.



Supplementary Fig. S20 | Comparison of CO adsorption ability of Pt species through CO reaction order. CO kinetic orders of 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) at 220 °C. The error bars are defined as standard deviation, and the center of each error bar represents the mean value of three independent experiments.

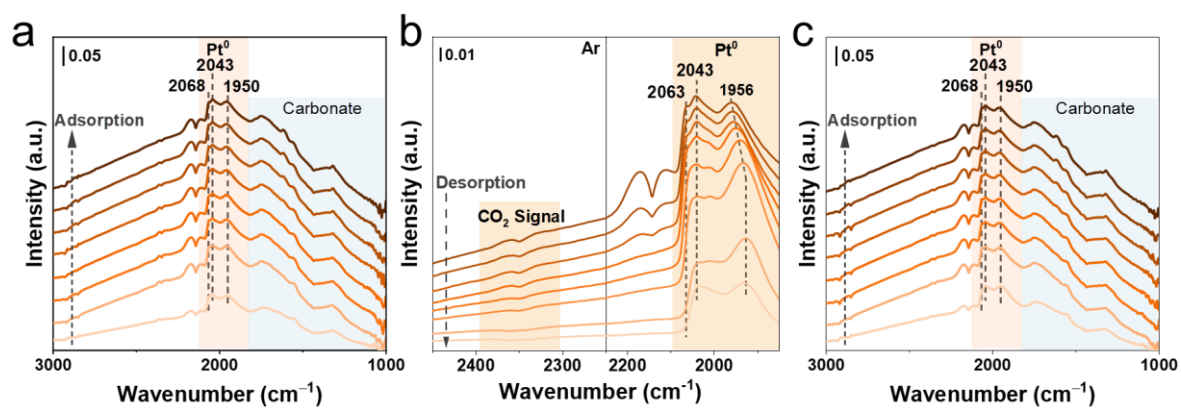


Supplementary Fig. S21 | Repeated CO injections on the removal of surface oxygen from the 0.3Pt/(TiO₂-A) catalyst. Four cycles of CO adsorption experiments conducted under *in situ* DRIFTS spectroscopy at 250 °C.

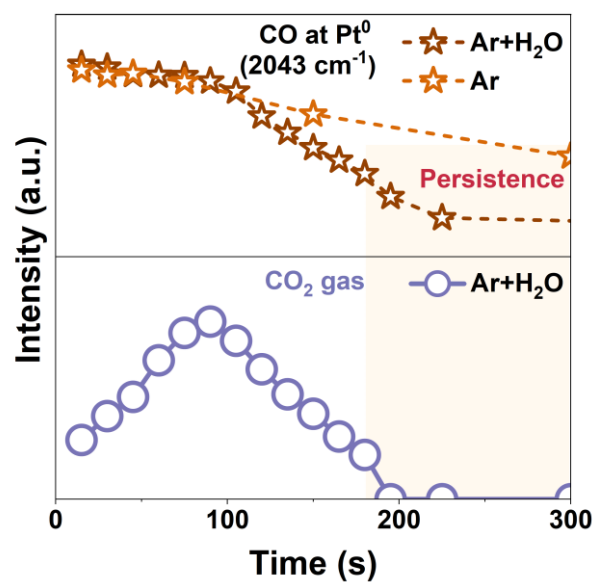


Supplementary Fig. S22 | CO adsorption-desorption behavior in 0.3Pt/(TiO₂-A) catalyst.

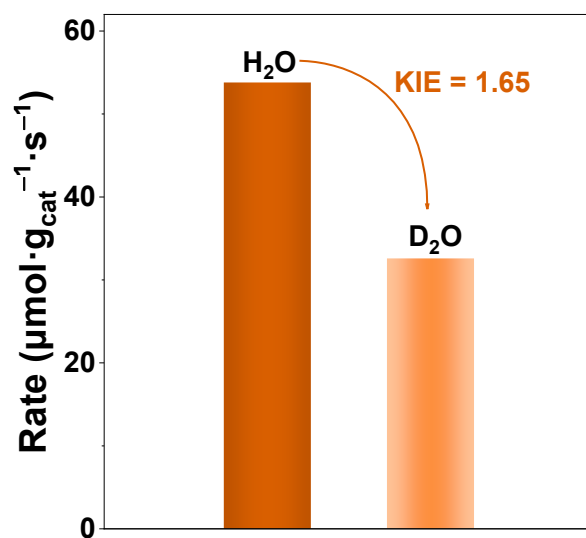
In situ DRIFTS spectra of (a) CO adsorption, (b) Ar desorption and (c) CO re-adsorption over 0.3Pt/(TiO₂-A) catalyst at 250 °C.



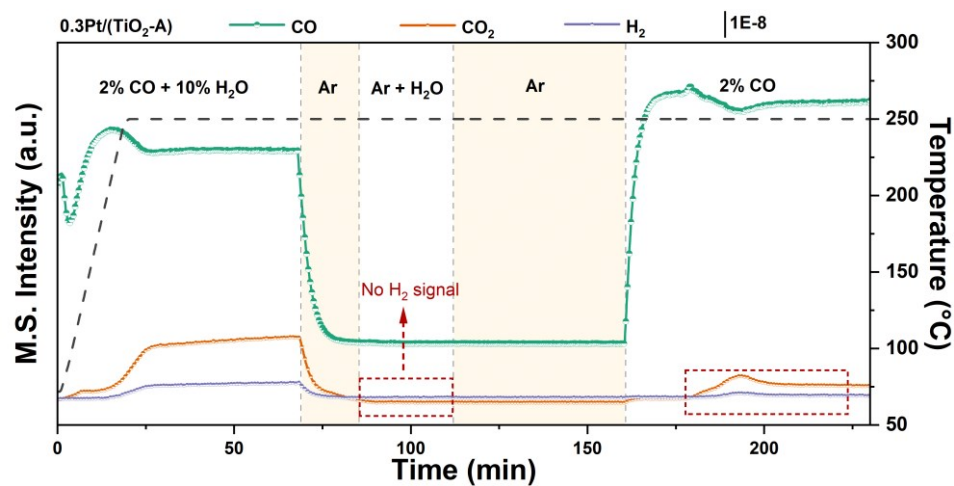
Supplementary Fig. S23 | CO adsorption-desorption behavior in 0.3Pt/(TiO₂-T-500) catalyst. *In situ* DRIFTS spectra of (a) CO adsorption, (b) Ar desorption and (c) CO re-adsorption over 0.3Pt/(TiO₂-T-500) catalyst at 250 °C.



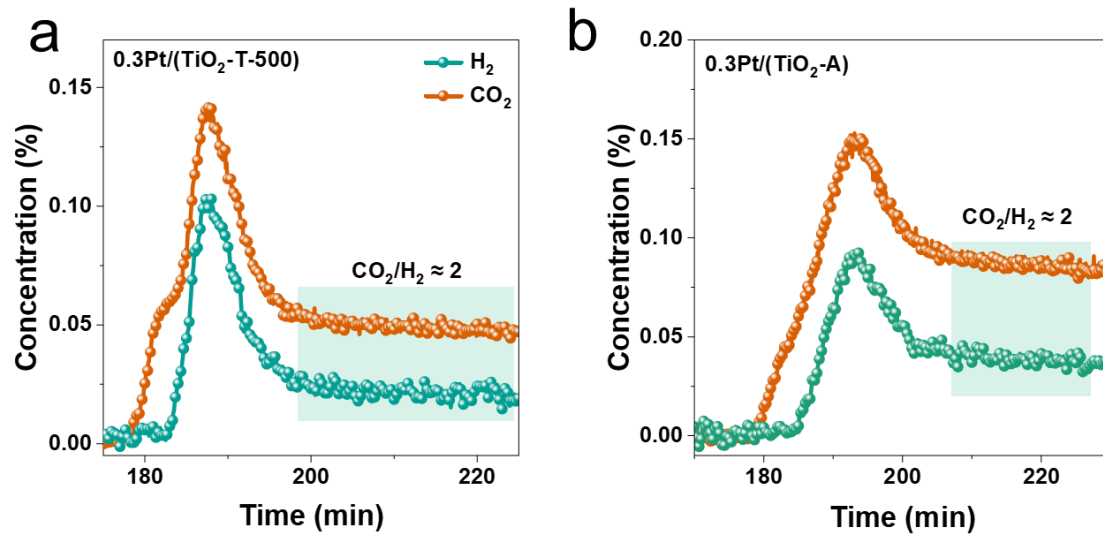
Supplementary Fig. S24 | Identification of active Pt sites. The intensity of CO adsorption at 2043 cm⁻¹ upon Ar+H₂O purging as function of time over 0.3Pt/(TiO₂-T-500).



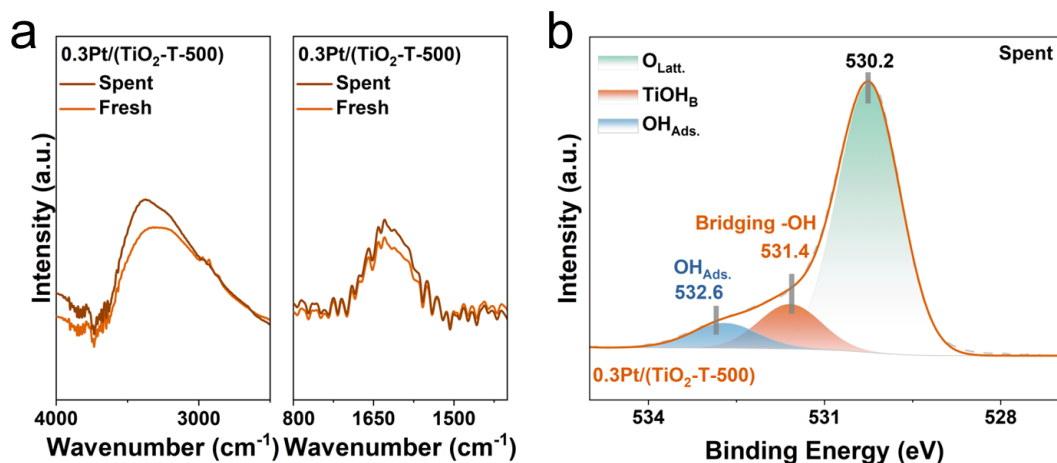
Supplementary Fig. S26 | Confirmation of H₂O dissociation as the rate-determining step in the WGS reaction. The H/D kinetic isotope effect (KIE) of 0.3Pt/(TiO₂-T-500) catalysts at 250 °C.



Supplementary Fig. S27 | Reaction mechanism of 0.3Pt/(TiO₂-A) catalyst. *In situ* WGS reaction, H₂O dissociation and CO-TPSR of the 0.3Pt/(TiO₂-A) catalyst.

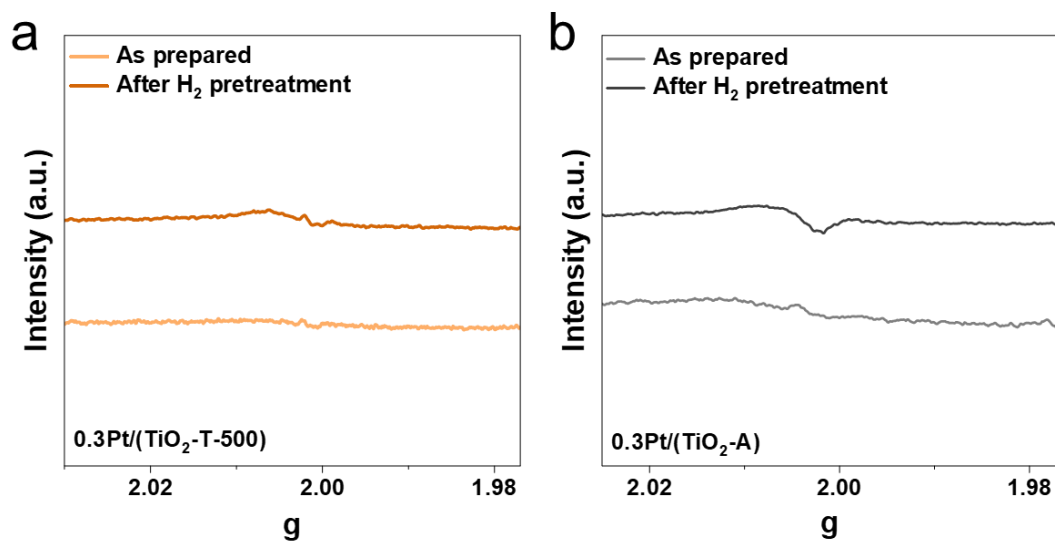


Supplementary Fig. S28 | Reaction mechanism of $0.3\text{Pt}/(\text{TiO}_2\text{-T-500})$ and $0.3\text{Pt}/(\text{TiO}_2\text{-A})$ catalyst. The concentration of CO_2 and H_2 during CO injection in the TPSR experiments of (a) $0.3\text{Pt}/(\text{TiO}_2\text{-T-500})$ and (b) $0.3\text{Pt}/(\text{TiO}_2\text{-A})$ catalyst at 250 °C.



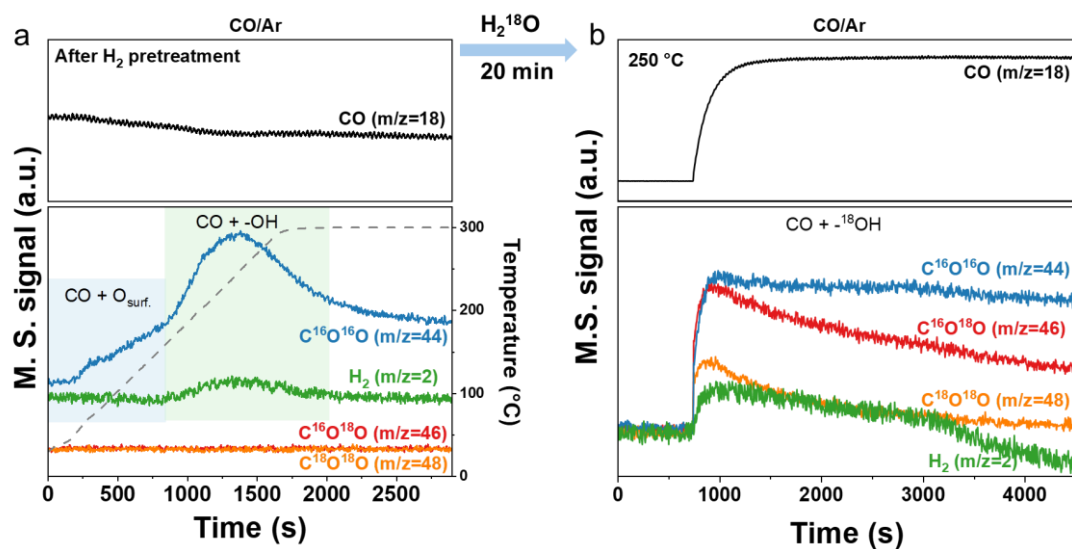
Supplementary Fig. S29 | Regeneration of surface hydroxyls on the spent 0.3Pt/(TiO₂-T-500) catalyst. (a) The attenuated total reflectance-infrared spectroscopy (ATR-IR) of the fresh and spent 0.3Pt/(TiO₂-T-500) catalyst. (b) O 1s XPS spectra of the spent 0.3Pt/(TiO₂-T-500) catalyst.

In the ATR-IR tests, the hydroxyl signal of the 0.3Pt/(TiO₂-T-500) catalyst significantly increased after the reaction compared to that before the reaction (Supplementary Fig. S29a). Additionally, the O 1s XPS spectra after the reaction exhibited supplementary signals corresponding to adsorbed hydroxyl groups, which was attributed to the adsorption of H₂O generated during the reaction on the catalyst surface. Concurrently, the peak area ratio of bridging hydroxyl was markedly enhanced compared to that on the support, which demonstrated the replenishment of bridging hydroxyl groups resulting from the dissociation of H₂O at oxygen vacancies on the TiO₂ surface (Supplementary Fig. S29b).



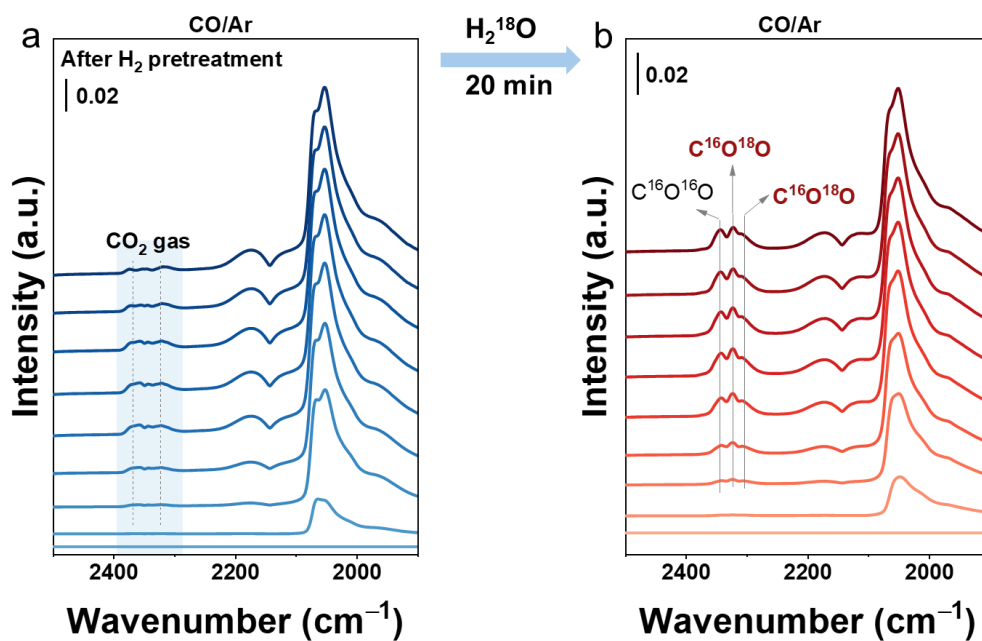
Supplementary Fig. S30 | Comparison of defect concentrations. EPR measurement of (a) 0.3Pt/(TiO₂-T-500) and (b) 0.3Pt/(TiO₂-A) before and after H₂ pretreatment.

The EPR results manifested that the 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts did not exhibit significant defects after H₂ pretreatment.

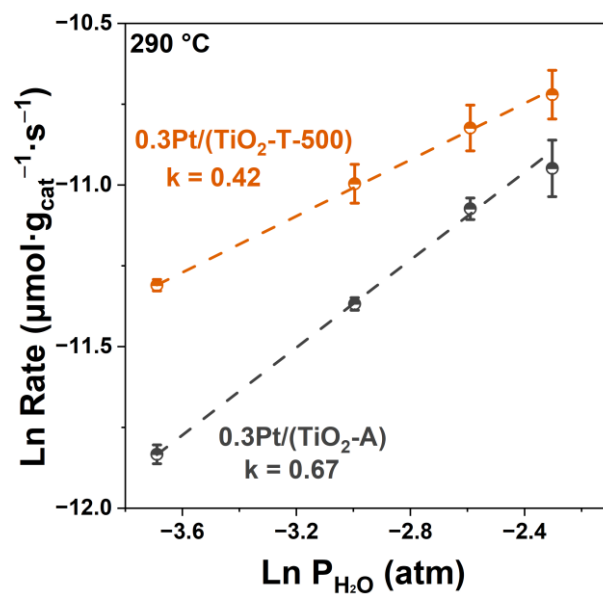


Supplementary Fig. S31 | Isotopic tracing of water dissociation using mass spectrometry.

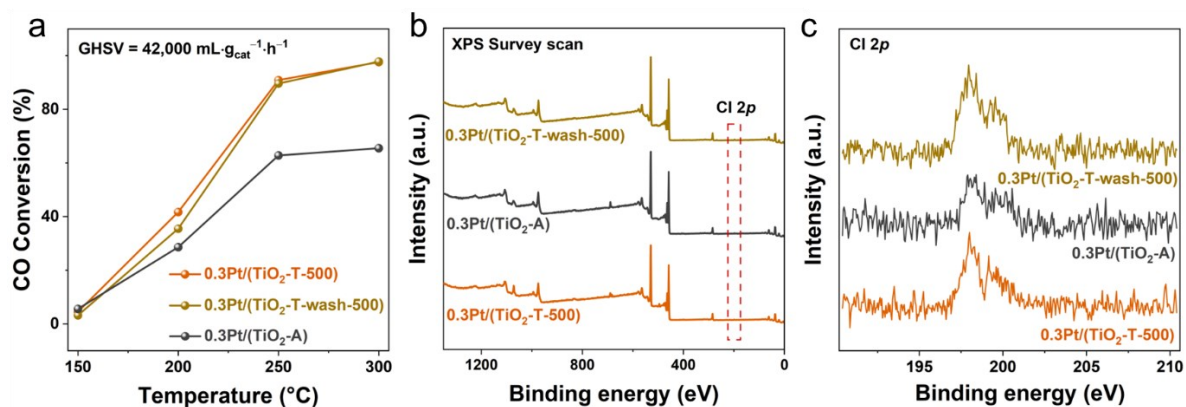
(a) CO-TPR profiles of 0.3Pt/(TiO₂-T-500) after H₂ pretreatment. (b) The generation of CO₂ when injecting CO/Ar after H₂¹⁸O dissociation at 250 °C.



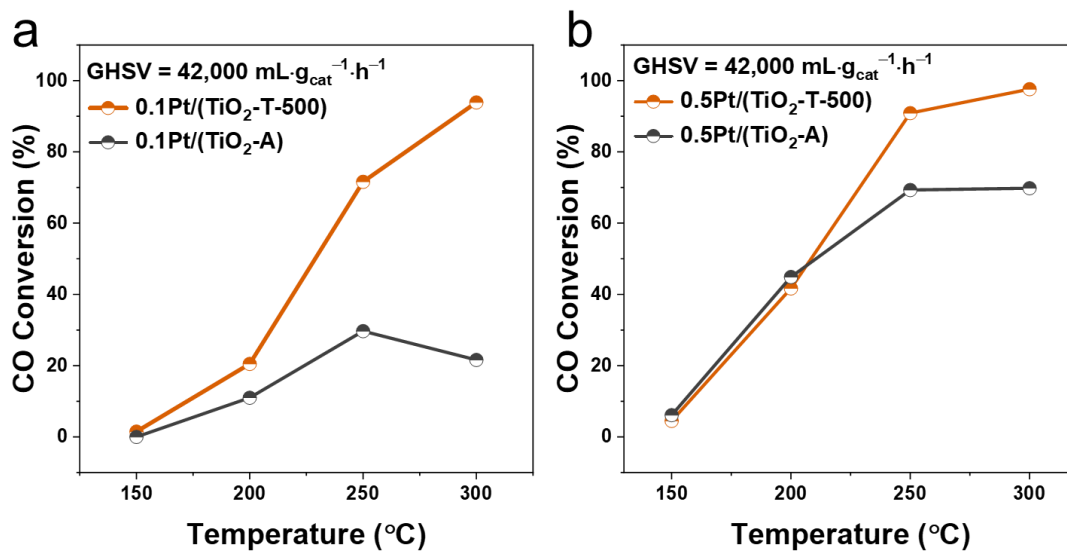
Supplementary Fig. S32 | Isotopic tracing of water dissociation using *in situ* DRIFTS spectra. *In situ* DRIFTS spectra of the CO adsorption for 0.3Pt/(TiO₂-T-500) after (a) H₂ pretreatment and (b) subsequent H₂¹⁸O treatment.



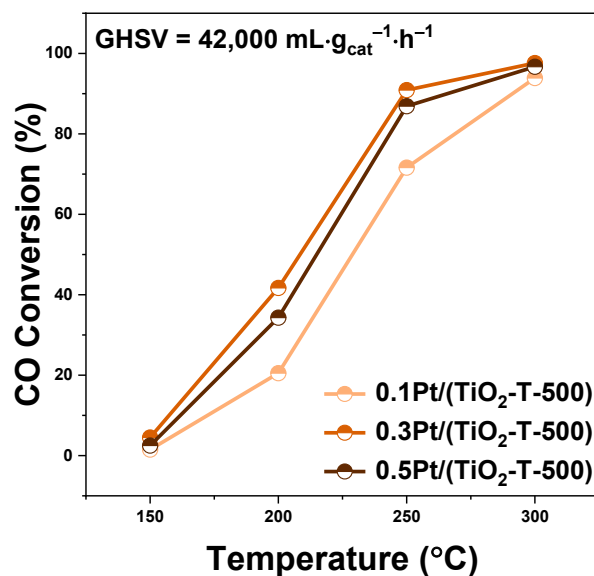
Supplementary Fig. S33 | Comparison of H₂O dissociation ability through H₂O reaction order. H₂O kinetic orders of 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) at 290 °C. The error bars are defined as standard deviation, and the center of each error bar represents the mean value of three independent experiments.



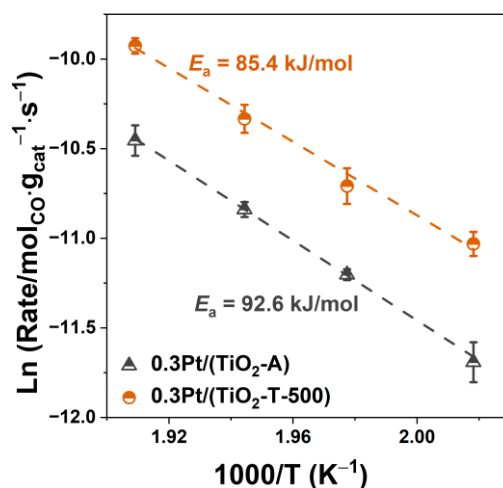
Supplementary Fig. S34 | Exclusion of residual Cl effects on catalytic performance. (a) Temperature-dependent CO conversion of 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-wash-500) catalysts for the WGS reaction. (b) The XPS survey and (c) Cl 2p spectra of the fresh 0.3Pt/(TiO₂-T-500), 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-wash-500) catalysts.



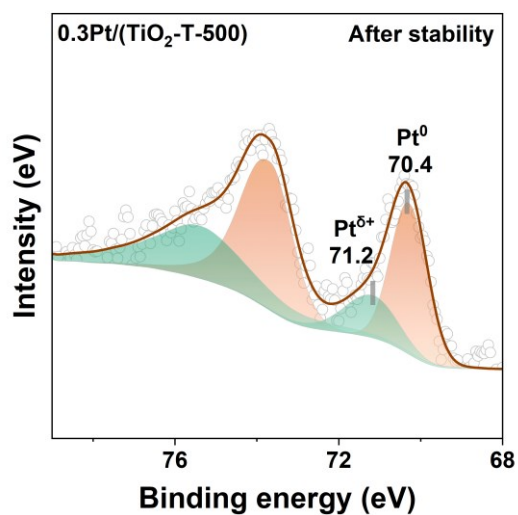
Supplementary Fig. S35 | The enhanced catalytic performance of Pt/(TiO₂-T-500) catalysts with different Pt loadings compared to Pt/(TiO₂-A) catalyst. Temperature-dependent CO conversion of Pt/(TiO₂-T-500) and Pt/(TiO₂-A) catalysts with (a) 0.1 % and (b) 0.5 % loading for the WGS reaction.



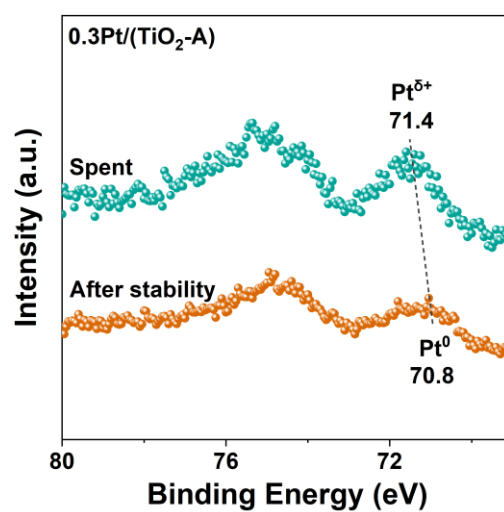
Supplementary Fig. S36 | The catalytic performance of Pt/(TiO₂-T-500) catalysts with various Pt loadings. Temperature-dependent CO conversion over Pt/(TiO₂-T-500) with various Pt loading under GHSV = 42,000 mL·g_{cat}⁻¹·h⁻¹. Reaction condition: 2% CO/10% H₂O/N₂.



Supplementary Fig. S37 | Arrhenius plots for the catalysts of 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) catalysts. Activation energy (E_a) for 0.3Pt/(TiO₂-A) and 0.3Pt/(TiO₂-T-500) catalysts. The error bars are defined as standard deviation, and the center of each error bar represents the mean value of three independent experiments.



Supplementary Fig. S38 | The electronic structure of Pt species in 0.3Pt/(TiO₂-T-500). Pt 4f XPS spectra of Pt species in 0.3Pt/(TiO₂-T-500) after long-time reaction for 40 h.



Supplementary Fig. S39 | The electronic structure of Pt species in 0.3Pt/(TiO₂-A). Pt 4f
XPS spectra of Pt species in 0.3Pt/(TiO₂-A) after long-time reaction for 40 h.

Supplementary Tables:

Supplementary Table 1. The CO uptake and Pt dispersion of the spent 0.3Pt/(TiO₂-A), 0.3Pt/(TiO₂-T-UC) and 0.3Pt/(TiO₂-T-500) catalysts.

	CO uptake ($\mu\text{mol/g}$)	Dispersion
0.3Pt/(TiO ₂ -A)	7.4	48.1 %
0.3Pt/(TiO ₂ -T-UC)	2.9	18.8 %
0.3Pt/(TiO ₂ -T-500)	8.5	55.2 %

Supplementary Table 3. Residual Na and Cl content on the catalyst surface.

Samples	Sodium residue (wt.%) ^a	Chloride residue (wt.%) ^b
0.3Pt/(TiO ₂ -T-500)	0.49	0.95
0.3Pt/(TiO ₂ -A)	<LOD	0.81
0.3Pt/(TiO ₂ -T-500)-wash	<LOD	1.07
0.3Pt/(TiO ₂ -T-400)	0.59	—

^a Determined by ICP-MS; ^b Determined by XPS; LOD: limit of detection. The method LOD was 0.00125 wt.% for Na.

For the analysis of Na residues, ICP-MS was employed to showed that 0.3Pt/(TiO₂-T-500) catalyst contained 0.49 wt.% Na. In contrast, the Na content in the 0.3Pt/(TiO₂-A) catalyst was below the detection limit (<50 ppt, 0.00125 wt.%). To rule out the role of residual Na, the activity of the 0.3Pt/(TiO₂-T-wash-500) catalyst, whose support was washed with 5 L of high-purity water to reduce Na content below the ICP-MS detection limit, was found to be comparable to that of 0.3Pt/(TiO₂-T-500) and higher than that of 0.3Pt/(TiO₂-A) (Supplementary Fig. 34a). This provided the understanding that the activity enhancement was not attributable to Na residues. Furthermore, we also found that the residual Na content on the surface of the 0.3Pt/(TiO₂-T-400) catalyst was at the same level as that on the 0.3Pt/(TiO₂-T-500) catalyst, but the catalytic reaction rate of 0.3Pt/(TiO₂-T-400) was only half of the 0.3Pt/(TiO₂-T-500) catalyst (Figure 1c). This further indicated that residual Na was not the source of the enhancement in catalytic performance.

For the analysis of Cl residues, we clarify that standard ICP-OES/MS is not the optimal technique for accurate quantification of chloride anions due to poor sensitivity in the vacuum UV region and significant spectral interferences, which preclude reliable detection at the low ppm/ppb levels relevant here. Thus, we could only perform a semi-quantitative assessment and trend analysis of the Cl residues on the surfaces of different catalysts using Cl 2p XPS. As indicated by semi-quantitative XPS analysis, the difference in Cl content between 0.3Pt/(TiO₂-T-500) and 0.3Pt/(TiO₂-A) was within 0.2 wt.%. Therefore, the observed activity difference between the two catalysts could not be attributed to residual Cl.

Supplementary Table 4. Comparison of catalytic performances for WGS reaction over various catalysts.

No.	Sample	Condition	Temperature (°C)	Rate ($\mu\text{mol}_{\text{CO}} \cdot \text{g}_{\text{cat}}^{-1} \cdot \text{s}^{-1}$)	Rate ($\mu\text{mol}_{\text{CO}} \cdot \text{g}_{\text{metal}}^{-1} \cdot \text{s}^{-1}$)	Ref.
1	1% Pt-CeO ₂ PD	2% CO/10% H ₂ O/N ₂	250 °C	33.3	3.3×10^3	6
2	1% Pt-CeO ₂ IMP	2% CO/10% H ₂ O/N ₂	250 °C	23.3	2.3×10^3	6
3	0.72% Pt(S-NPs)/CeO ₂	2% CO/5% H ₂ O/N ₂	250 °C	15.4	2.1×10^3	7
4	2.13% Pt/FeCeAl	4.5% CO/30% H ₂ O/N ₂	250 °C	3.3	1.5×10^2	8
5	1% PtNa-SiO ₂	2% CO/10% H ₂ O/He	250 °C	~6.0	6×10^2	9
6	0.05% Pt-FeO _x	2% CO/10% H ₂ O/He	250 °C	2.4	4.8×10^3	10
7	0.5% Pt-Na-TiO ₂	3% CO/10% H ₂ O/87% He	250 °C	38.4	7.7×10^3	11
8	0.5% Pt-TiO ₂	3% CO/10% H ₂ O/87% He	250 °C	10.3	2.1×10^3	11
9	0.5% Pt-TiO ₂	3% CO/10% H ₂ O/He	200 °C	2.3	4.6×10^2	12
10	1.72% Pt-mpTiO ₂ -PS ₁₂₀	2% CO/8% H ₂ O/N ₂	200 °C	23.8	1.4×10^3	13
11	1.26% Pt-npTiO ₂	2% CO/8% H ₂ O/N ₂	200 °C	2.6	2.0×10^2	13
12	2% Au @TiO ₂ -x/ZnO(H300)	6% CO/25% H ₂ O/Ar	250 °C	15.2	7.6×10^2	14
13	2% Au @TiO ₂ -x/ZnO(H300)	6% CO/25% H ₂ O/Ar	200 °C	8.6	4.3×10^2	14
14	0.5% Pt/Ce _{0.8} Ti _{0.2} O _{2-δ}	3% CO/10% H ₂ O/He	200 °C	0.78	1.6×10^2	15
15	1% Au-Ti(15Ce)O ₂	1% CO/3% H ₂ O/He	225 °C	3.1	3.1×10^2	16
16	0.5% Pt/3Ce-10Ti	2% CO/10% H ₂ O/N ₂	200 °C	2.0	4.1×10^2	17
17	0.5% Pt/3Ce-10Ti	2% CO/10% H ₂ O/N ₂	225 °C	5.2	1.0×10^3	17
This work	0.3% Pt/(TiO₂-T-500)	2% CO/10% H₂O/N₂	200 °C	2.8	9.3×10^2	—
This work	0.3% Pt/(TiO₂-T-500)	2% CO/10% H₂O/N₂	225 °C	13.5	4.5×10^3	—
This work	0.3% Pt/(TiO₂-T-500)	2% CO/10% H₂O/N₂	250 °C	52.3	1.7×10^4	—
This work	0.3% Pt/(TiO₂-T-UC)	2% CO/10% H₂O/N₂	250 °C	4.9	1.6×10^3	—
This work	0.3% Pt/(TiO₂-A)	2% CO/10% H₂O/N₂	250 °C	26.2	5.2×10^3	—
This work	0.3% Pt/CeO₂	2% CO/10% H₂O/N₂	250 °C	10.4	3.5×10^3	—

Supplementary Table 5. Comparison of reaction rate and TOF values for WGS reaction over various catalysts at 250 °C.

Samples	Rate (mol _{CO} ·g _{Pt} ⁻¹ ·h ⁻¹)	TOF (s ⁻¹)	Reference
0.3Pt/(TiO ₂ -T-500)	62.7	6.15	This work
0.3Pt/(TiO ₂ -A)	31.4	3.54	This work
Pt/TiO ₂	7.42	0.46	11
Pt/0.06Na/TiO ₂	27.6	1.58	
Pt-4Na/TiO ₂	7.74	1.39	18
Pt/FeO _x	17.2	0.93	10
Pt/Al ₂ O ₃	0.35	0.02	19
Pt/CeO _x	1.71	0.17	20
Pt=Na/SiO ₂	4.32	0.24	21

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